MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, SILICON, MULTIPLE DIODE ARRAYS,
TYPES 1N5768, 1N5770, 1N5772, 1N5774, 1N6100, 1N6101, 1N6496,
1N6506, 1N6507, 1N6508, 1N6509, 1N6510, AND 1N6511
JAN, JANTX, JANTXV, AND JANS

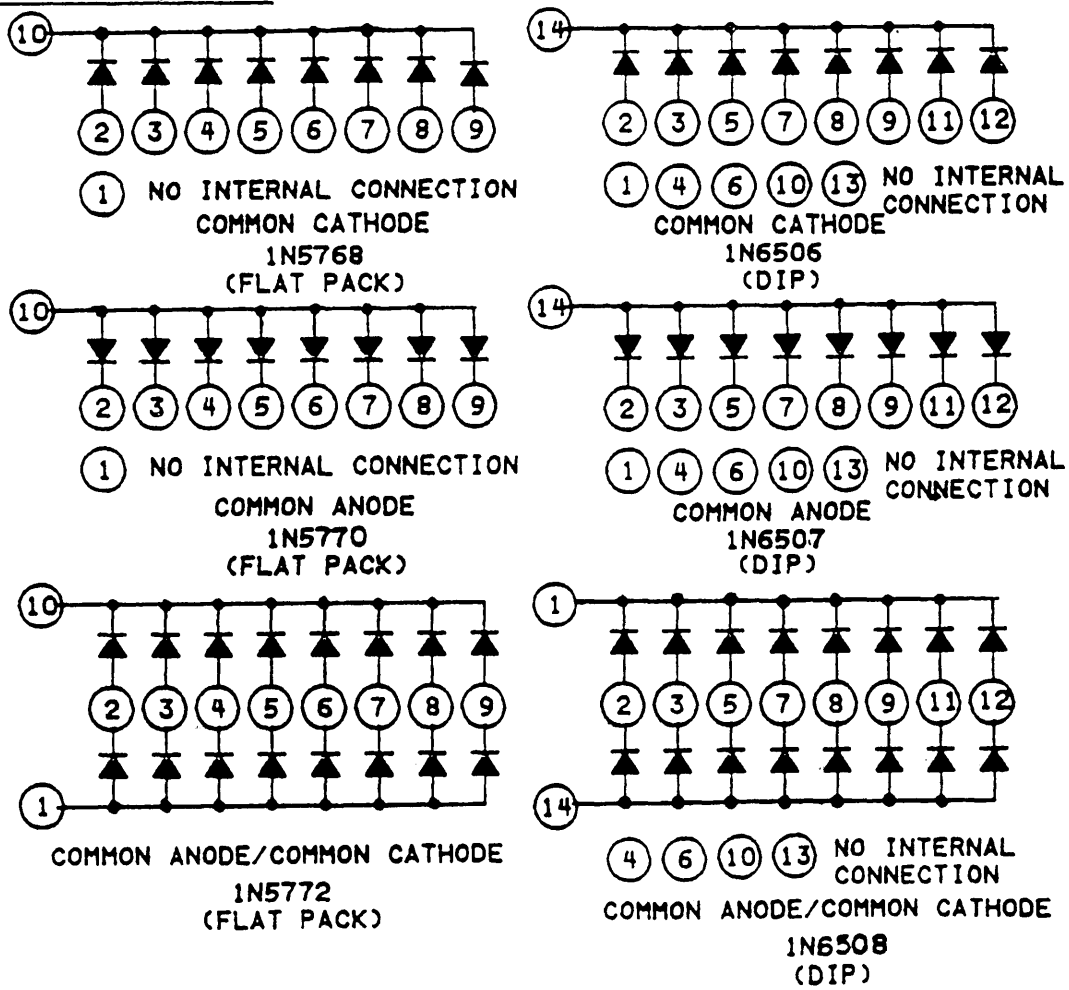
This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon, multiple diode arrays. Four levels of product assurance are provided for each device type as specified in MIL-S-19500.

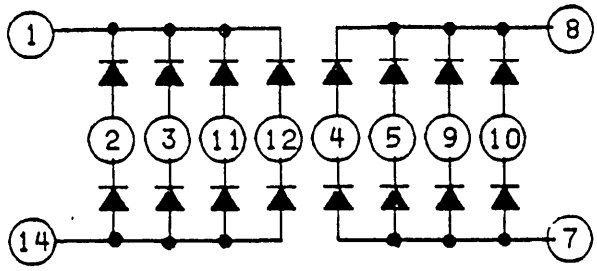
1.2 Physical dimensions. See figures 1, 2, 3, 4, 5, and 6.

1.3 Schematic configurations.



Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, ATTN: RADC/RBE-2, Griffiss AFB, NY 13441-5700, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

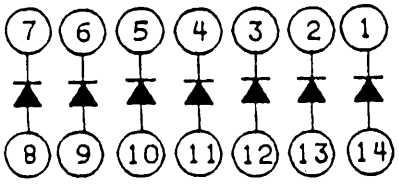
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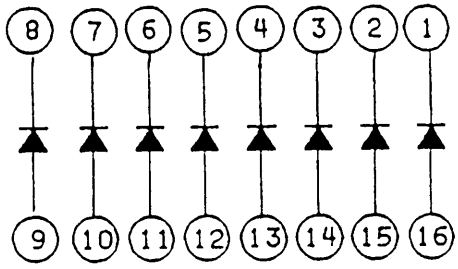
13 6 NO INTERNAL CONNECTION

DUAL COMMON ANODE/COMMON CATHODE

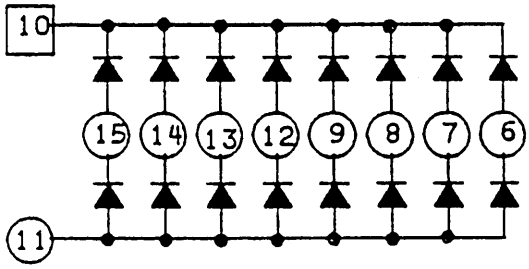
1N5774 (FLAT PACK) 1N6509 (DIP)

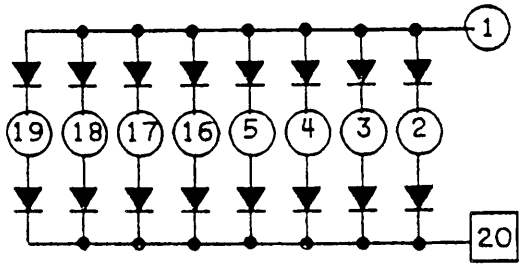


1N6100 (FLAT PACK) 1N6511 (DIP)



1N6101 (DIP) 1N6510 (FLAT PACK)





DUAL COMMON ANODE/COMMON CATHODE

1N6496 (FLAT PACK)

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1.4 Maximum ratings.

Type	$V_{BR(R)}$ 1/2/	I_O 1/3/ T_A = +25°C	I_{FSM} 1/ t_p = 1/120 s	P_T 4/ T_A = +25°C	T_{op}	T_{STG}
	V dc	mA dc	mA dc	mW	°C	°C
	60	300	500		-65 to +150	-65 to +200
1N5768				500		
1N5770				500		
1N5772				500		
1N5774				500		
1N6496				500		
1N6506				600		
1N6507				600		
1N6508				600		
1N6509				600		
1N6100	75			500		
1N6101				600		
1N6510				500		
1N6511				600		

1/ Each diode.
2/ Pulsed: PW = 100 ms maximum; duty cycle ≤ 20%.
3/ Derate at 2.4 mA/°C above +25°C.
4/ Derate at 4.0 mW/°C above +25°C.

1.5 Electrical characteristics, each diode.

Type	V_{F1} I_F = 100 mA dc		V_{F1} 1/ I_F = 500 mA dc		I_{R1} V_R = 40 V dc		C_t		t_{fr} I_F = 500 mA dc		t_{rr} I_F = I_R = 200 mA dc R_L = 100Ω i_{rr} = 20 mA dc	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
	---	1.0 V dc	---	1.5 V dc	---	0.1 µA dc			---	40 ns	---	20 ns
1N5768								4 pF				
1N5770								8 pF				
1N5772								8 pF				
1N5774								8 pF				
1N6496								8 pF				
1N6506								4 pF				
1N6507								8 pF				
1N6508								8 pF				
1N6509								8 pF				

Type	V_{F1} I_F = 100 mA dc		I_{R1} V_R = 20 V dc		C_t		t_{fr} I_F = 100 mA dc		t_{rr} I_F = I_R = 10 mA dc R_L = 100Ω i_{rr} = 1 mA dc	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
	---	1.0 V dc	---	25 nA dc	---	4 pF	---	15 ns	---	5 ns
1N6100										
1N6101										
1N6510										
1N6511										

1/ Pulsed: PW = 300 µs, duty cycle ≤ 2%, 90 µs after leading edge of pulse.

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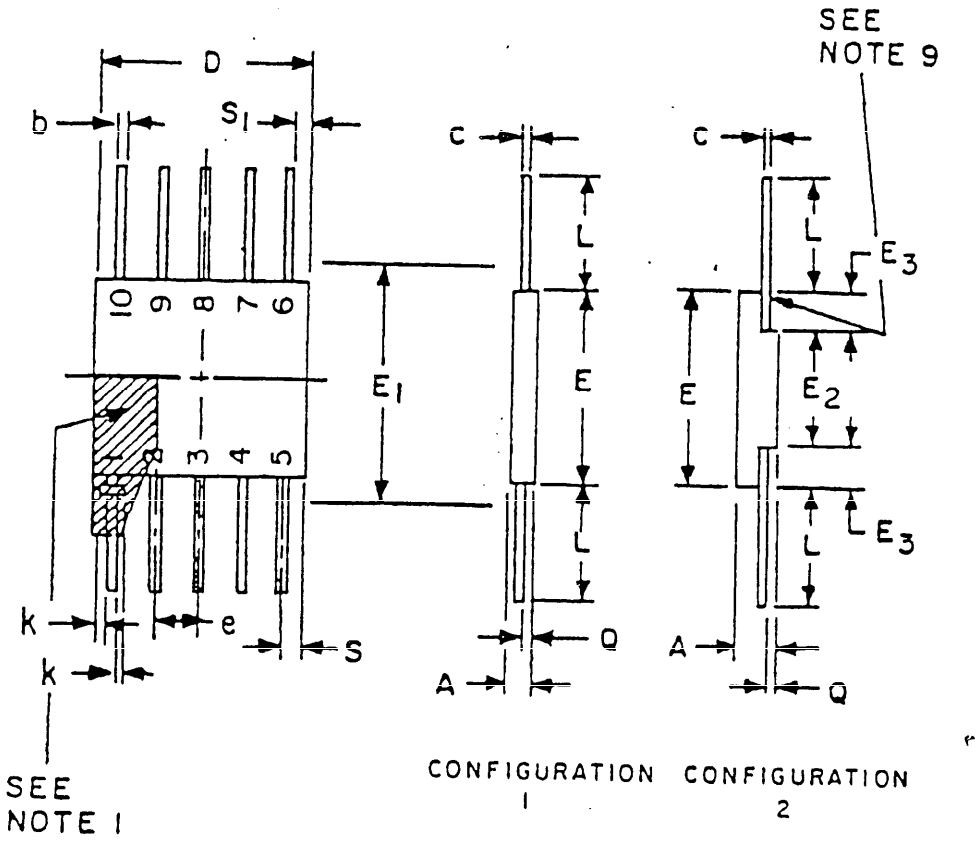


FIGURE 1. Physical dimensions for types 1N5768, 1N5770, and 1N5772.

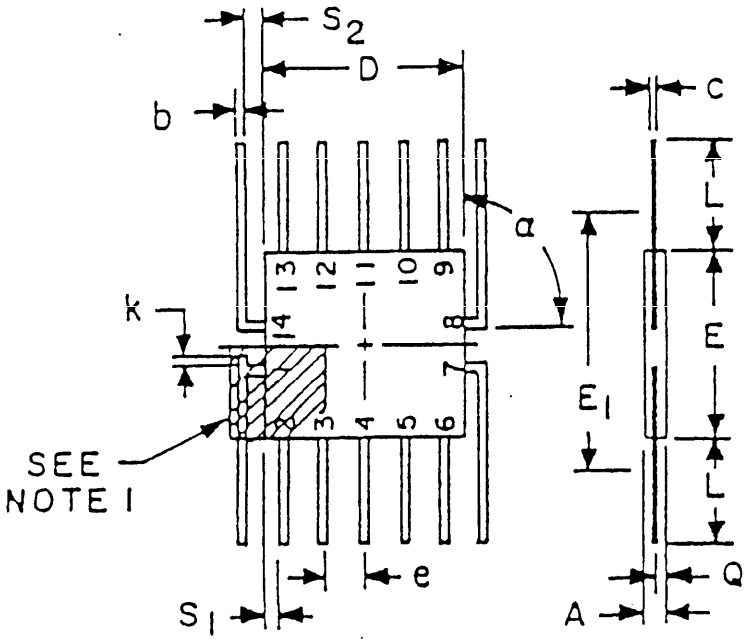
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Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max	
A	.030	.090	0.76	2.29	
b	.010	.019	0.25	0.48	5
c	.003	.006	0.08	0.15	5
D		.290		7.37	3
E	.240	.260	6.10	6.60	
E_1		.280		7.11	3
E_2	.125		3.18		
E_3	.030		0.76		
e	.050 BSC		1.27 BSC		4, 6
k	.008	.015	0.20	0.38	10
L	.240	.370	6.10	9.40	
Q	.005	.050	0.13	1.27	2
S		.045		1.14	7
S_1	.005		0.13		7, 8

NOTES:
1. Index area: A notch or a pin one identification mark shall be located adjacent to pin one and shall be within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark. Alternatively, a tab (dim. k) may be used to identify pin one. This tab may be located on either side as shown.
2. Dimensions Q shall be measured at the point of exit of the lead from the body. Dimension Q shall be .0085 (0.22 mm) minimum when lead finish A is applied.
3. These dimensions allows for off-center lid, meniscus, and glass overrun.
4. The basic pin spacing is .050 (1.27 mm) between centerlines. Each pin centerline shall be located within ±.005 (0.13 mm) of its exact longitudinal position relative to pins 1 and 14.
5. All leads: Increase maximum limit by .003 (0.08 mm) measured at the center of the flat, when lead finish A is applied.
6. Eight spaces.
7. Applies to all four corners (lead numbers 1, 5, 6, and 10).
8. Dimensions S_1 may be .000 (.00 mm) if lead numbers 1, 5, 6, and 10 bend toward the cavity of the package within one lead's width from the point of entry of the lead into the body. For all bottom-brazed or side-brazed configurations, dimension S_1 shall be measured from the edge of the furthest extension of the metal pad or lead.
9. Optional configuration. If this configuration is used, no organic or polymeric materials shall be molded to the bottom of the package to cover the leads.
10. Optional, see note 1. If a pin one identification mark is used in addition to this tab, the minimum limit of dimension k does not apply.

FIGURE 1. Physical dimensions for types 1N5768, 1N5770, and 1N5772.

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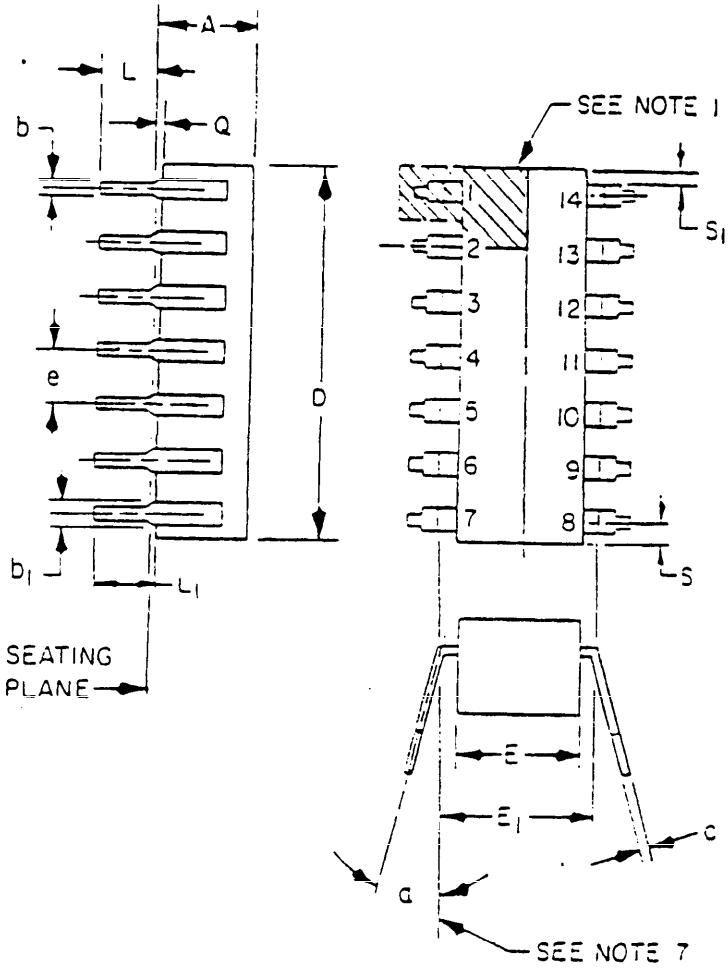
Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max	
A	.030	.085	0.76	2.16	
b	.010	.019	0.25	0.48	5
c	.003	.006	0.08	0.15	5
D		.280		7.11	3
E	.240	.260	6.10	6.60	
E_1		.280		7.11	3
e	.050 BSC		1.27 BSC		4, 6
k	.008	.015	0.20	0.38	9
L	.250	.370	6.35	9.40	
Q	.010	.040	0.25	1.02	2
S_1	.005		0.13		7, 8
S_2	.004		0.10		10
α	30°	90°	30°	90°	11

NOTES:
1. Index area: A notch or a pin one identification mark shall be located adjacent to pin one and shall be within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark. Alternatively, a tab (dim. k) may be used to identify pin one. This tab may be located on either side as shown.
2. Dimensions Q shall be measured at the point of exit of the lead from the body. Dimension Q shall be be .0085 (0.22 mm) minimum when lead finish A is applied.
3. These dimensions allows for off-center lid, meniscus, and glass overrun.
4. The basic pin spacing is .050 (1.27 mm) between centerlines. Each pin centerline shall be located within ±.005 (0.13 mm) of its exact longitudinal position relative to pins 1 and 14.
5. All leads: Increase maximum limit by .003 (0.08 mm) measured at the center of the flat, when lead finish A is applied.
6. Twelve spaces.
7. Applies to all four corners (lead numbers 2, 6, 9, and 13).
8. Dimensions S_1 may be .000 (.00 mm) if lead numbers 2, 6, 9, and 13 bend toward the cavity of the package within one leads's width from the point of entry of the lead into the body.
9. Optional, see note 1. If a pin one identification mark is used in addition to this tab, the minimum limit of dimensions k does not apply.
10. Applies to lead numbers 1, 7, 8, and 14.
11. Lead configuration is optional within dimension E except dimensions b and c apply.

FIGURE 2. Physical dimensions for types 1N5774 and 1N6100.

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Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max	
A		.200		5.08	
b	.014	.023	0.36	0.58	8
b_1	.030	.070	0.76	1.78	2, 8
c	.008	.015	0.20	0.38	8
D		.785		19.94	4
E	.220	.310	5.59	7.87	4
E_1	.290	.320	7.37	8.13	7
e	.100 BSC.		2.54 BSC		5, 9
L	.125	.200	3.18	5.08	
L_1	.150		3.81		
Q	.015	.060	0.38	1.52	3
S		.098		2.49	6
S_1	.005		0.13		6
α	0°	15°	0°	15°	

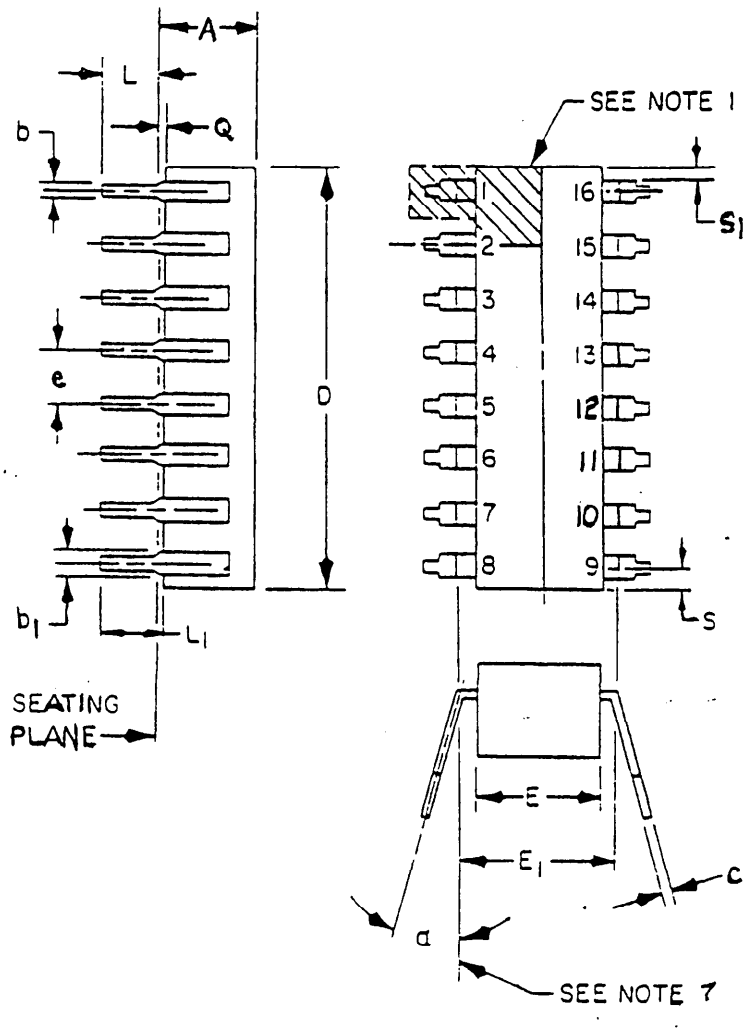


NOTES:
1. Index area: A notch or a pin one identification mark shall be located adjacent to pin one and shall be within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark.
2. The minimum limit for dimension b_1 may be .023 (0.58 mm) for lead numbers 1, 7, 8, and 14 only.
3. Dimension Q shall be measured form the seating plane to the base plane.
4. This dimension allows for off-center lid, meniscus, and glass overun.
5. The basic pin spacing is .100 (2.54 mm) between centerlines. Each pin centerline shall be located within ±.010 (0.25 mm) of its exact longitudinal position relative to pins 1 and 14.
6. Applies to all four corners (lead numbers 1, 7, 8, and 14).
7. Lead center when α is 0°. E_1 shall be measured at the centerline of the leads.
8. All leads. Pointed or rounded leads ends are allowed.
9. Twelve spaces.
10. No organic or polymeric materials shall be molded to the bottom of the package to cover leads.

FIGURE 3. Physical dimensions for types 1N6506, 1N6507, 1N6508, 1N6509, and 1N6511.

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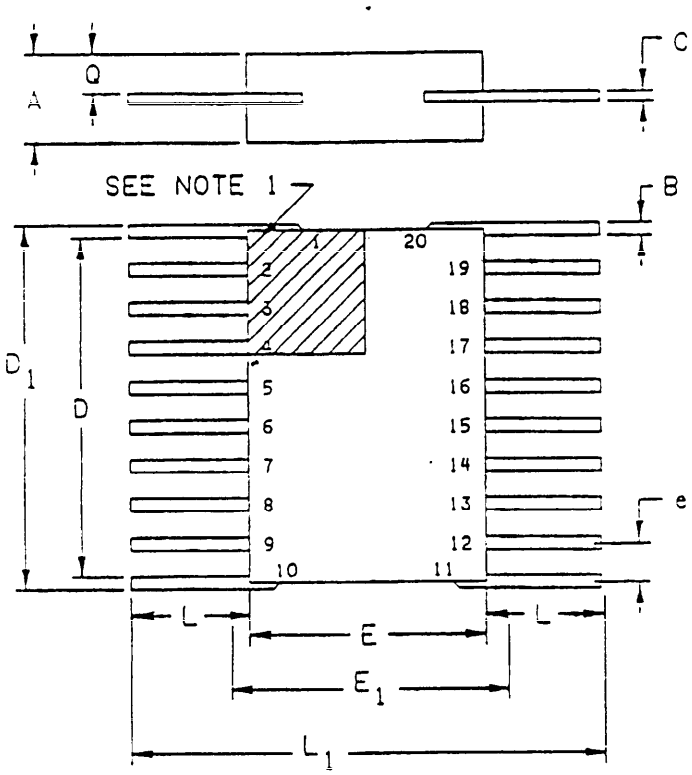
Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max	
A		.200		5.08	
b	.014	.023	0.36	0.58	8
b_1	.030	.070	0.76	1.78	2, 8
c	.008	.015	0.20	0.38	8
D		.785		19.94	4
E	.220	.310	5.59	7.87	4
E_1	.290	.320	7.37	8.13	7
e	.100 BSC		2.54 BSC		5, 9
L	.100	.200	2.54	5.08	
L_1	.150		3.81		
Q	.015	.060	0.38	1.52	3
S		.098		2.49	6
S_1	.005		0.13		6
α	0°	15°	0°	15°	



NOTES:
1. Index area: A notch or a pin one identification mark shall be located adjacent to pin one and shall be within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark.
2. The minimum limit for dimension b_1 may be .020 (0.51 mm) for lead numbers 1, 8, 9, and 16 only.
3. Dimension Q shall be measured form the seating plane to the base plane.
4. This dimension allows for off-center lid, meniscus, and glass overun.
5. The basic pin spacing is .100 (2.54 mm) between centerlines. Each pin centerline shall be located within ±.010 (0.25 mm) of its exact longitudinal position relative to pins 1 and 16.
6. Applies to all four corners (lead numbers 1, 8, 9, and 16).
7. Lead center when α is 0°. E_1 shall be measured at the centerline of the leads.
8. All leads: Increase maximum limit by .003 (0.08 mm) measured at the center of the flat, when lead finish A is applied. Pointed or rounded lead ends are allowed.
9. Fourteen spaces.

FIGURE 4. Physical dimensions for 1N6101.

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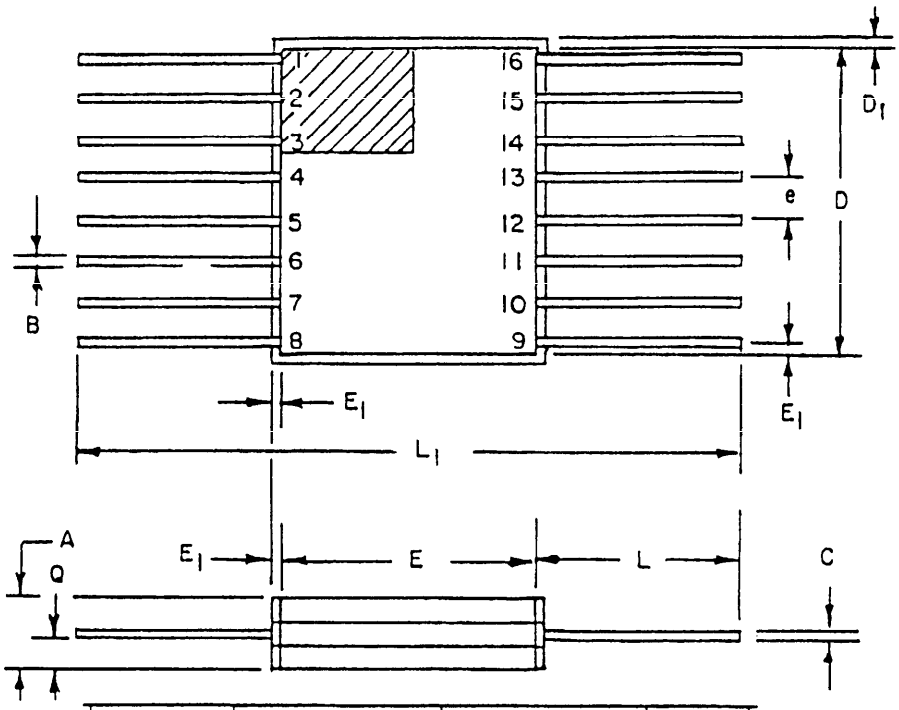


Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max	
A	.045	.075	1.14	1.91	
B	.010	.015	0.25	0.38	
C	.004	.006	0.10	0.15	
D	.272	.278	6.92	7.07	
D_1		.288		7.33	2
E	.194	.200	4.93	5.09	
E_1		.210		5.34	2
e	0.030 T.P.		0.76 T.P.		3, 5
L	0.265	0.290	6.74	7.38	
L_1	.724	.780	18.42	19.81	
Q	.020	.025	0.50	0.64	4

NOTES:
1. Index area: A notch or dot shall be located adjacent to pin 1 and shall be within shaded area limited by pin 3 and package center.
2. This dimension allows for off-center lid, meniscus, and glass overrun.
3. The true-position pin spacing is located within ±.005 inch (0.127 mm) of its true longitudinal position relative to pins 1 and 20.
4. Dimension "Q" shall be measured at the point of exit of the lead from the body.
5. Eighteen spaces.

FIGURE 5. Physical dimensions for 1N6496.

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Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max	
A	.060	.085	1.52	2.16	
B	.015	.019	0.38	0.48	
C	.003	.008	0.08	0.15	
D	.370	.400	9.40	10.16	
D_1		.020		0.51	2
E	.245	.260	6.22	6.60	
E_1		.015		0.38	2
e	.050 BSC		1.27 BSC		3, 5
L	.250	.370	6.35	9.40	
L_1	.745		18.92		
Q	.025	.040	0.64	1.01	

NOTES:
1. Index area: A notch or dot shall be located adjacent to pin 1 and shall be within shaded area limited by pin 3 and package center.
2. This dimension allows for off-center lid, menisus, and glass overrun.
3. The true-positon pin spacing is located within ±.005 inch (0.127 mm) of its true longitudinal position relative to pins 1 and 16.
4. Dimension "Q" shall be measured at the point of exit of the lead from the body.
5. Fourteen spaces.

FIGURE 6. Physical dimensions for 1N6510.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.
MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500, and as follows:

I_{RX} - Reverse current for each diode of test section.

V_{FX} - Forward voltage for each diode of test section.

I_{Ri} - Isolation current between any two interconnect pins of adjacent parallel sets of diodes with all other pins open circuited.

3.3 Design, construction, and physical dimensions. Diode arrays shall be of the design, construction, and physical dimensions shown on figures 1, 2, 3, 4, and 5.

3.3.1 Lead material and finish. Lead material shall be Kovar, F-15 alloy, or alloy 42 ASTM F-30. Lead finish shall be tinned or gold-plated or solder dipped. Where a choice of lead material of finish is desired, it shall be specified in the contract or purchase order (see 6.2).

3.4 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the diode array at the option of the manufacturer:

a. Country of origin.

b. Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

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4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANTX, JANTXV, and JANS levels only). Screening shall be in accordance with MIL-S-19500 (table II) and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurements	
	JANS level	JANTX and JANTXV levels
1	MIL-STD-883, method 2010, condition B.	MIL-STD-883, method 2010, condition B (JANTXV only).
9 and 10	Not applicable	Not applicable
11	I_{R1} and V_{F1}	I_{R1} and V_{F1}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{R1} = 100% of initial reading or 10 nA dc, whichever is greater. ΔV_{F1} = ±25 mV dc of initial reading.	Subgroups 2 of table I herein; ΔI_{R1} = 100% of initial reading or 10 nA dc, whichever is greater. ΔV_{F1} = ±25 mV dc of initial reading.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_A = 150°C. Time = 72 hours minimum.
V_R = 50 V dc for 1N5768, 1N5770, 1N5772, 1N5774, 1N6496, 1N6506, 1N6507, 1N6508, and 1N6509.
V_R = 60 V dc for 1N6100, 1N6101, 1N6510, and 1N6511.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JAN, JANTX, JANTXV) of MIL-S-19500, and tables IIa and IIb herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and table III, herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table IV herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Measurement of reverse current for 1N5772, 1N5774, 1N6496 1N6508 and 1N6509. For 1N5772, 1N5774, 1N6496, 1N6508, and 1N6509, the reverse current shall be measured using a circuit which bypasses the shunt resistance through the other diodes not under test, around the current meter. Care should be taken to assure that the voltage drop across the current meter is less than 10 millivolts.

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4.5.2 Forward voltage. This parameter shall be measured 90 microseconds after the leading edge of the pulse.

4.5.3 Peak forward voltage. During this test, the maximum shunt capacitance across the diode shall be 19 picofarads and the equipment bandwith shall be 80 megahertz minimum.

4.5.4 Pin-to-pin capacitance. This parameter is the total pin-to-pin capacitance across each individual diode and may not necessarily represent actual diode capacitance since they are all connected together at either the anode or cathode and these connections represent additional capacitance.

4.5.5 Reverse current (I_{RX}) and forward voltage (V_{FX}). Each common anode section and each common cathode section shall be tested separately. Each diode in the test section shall be measured individually after the array has reached thermal equilibrium.

4.5.6 Isolation current (I_R). These devices shall be subjected to the isolation current tests as specified:

a. For types 1N5772, 1N5774, 1N6496, 1N6508, and 1N6509 the isolation current shall be measured between any two interconnect pins of adjacent parallel sets of diodes with all other pins open-circuited. The interconnect pins are defined as pin numbers 2, 3, 4, 5, 6, 7, 8, and 9 for array 1N5772; pin numbers 2, 3, 4, 5, 9, 10, 11, and 12 for array 1N5774 and 1N6509; and pin numbers 2, 3, 4, 5, 6, 7, 8, 9, 12, 13, 14, 15, 16, 17, 18, and 19 for array 1N6496; and pin numbers 2, 3, 5, 7, 8, 9, 11, and 12 for array 1N6508. The isolation current shall be measured between all parallel sets of diodes.

b. For types 1N6100 and 1N6511, the isolation current shall be measured between all adjacent cathode pins (1 through 7) and between all adjacent anode pins (8 through 14). For type 1N6101, and 1N6510 the isolation current shall be measured between all adjacent cathode pins (1 through 8) and between all adjacent anode pins (9 through 16). More than one pair of diodes may be tested at one time but all of the anodes shall be open-circuited while the cathode leakage is being tested, and all of the cathodes shall be open-circuited when the anode leakage is being tested.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data. Acquisition documents should specify the lead material and finish.

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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TABLE I. Group A inspection.

Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	LTPD JANS 1/	LTPD JAN TX 1/ TXV	Symbol	Limits Min	Limits Max	Units
Subgroup 1								
Visual and mechanical inspection	2071				---	---	---	---
Subgroup 2								
Breakdown voltage	4021				$V_{(BR)R}$			
1N5768, 1N5770, 1N5772, 1N5774, 1N6496, 1N6506, 1N6507, 1N6508, 1N6509		I_R = 10 µA dc; PW = 100 ms; duty cycle ≤ 20%				60	---	V dc
1N6100, 1N6101, 1N6510, 1N6511		I_R = 5 µA dc; PW = 100 ms; duty cycle ≤ 20%				75	---	V dc
Reverse current	4016	DC method (see 4.5.1)						
All		V_R = 40 V dc			I_{R1}	---	0.1	µA dc
1N6100, 1N6101, 1N6510, 1N6511		V_R = 20 V dc			I_{R2}	---	25	nA dc
Forward voltage	4011	PW = 300 µs; duty cycle ≤ 2% (see 4.5.2)						
All		I_F = 100 mA dc			V_{F1}	---	1.0	V dc
1N5768, 1N5770, 1N5772, 1N5774, 1N6496, 1N6506, 1N6507, 1N6508, 1N6509		I_F = 500 mA dc			V_{F2}	---	1.5	V dc
Subgroup 3								
High temperature operation		T_A = 150°C						
Reverse current	4016	DC method; V_R = 40 V dc			I_{R3}	---	50	µA dc
Low temperature operation		T_A = -55°C						
Forward voltage	4011	I_F = 10 mA dc			V_{F3}	---	1	V dc

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See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD		Symbol	Limits		Units
	Method	Conditions	JANS 1/	JAN TX 1/ TXV		Min	Max	
Subgroup 4								
Forward recovery voltage (peak)	4026	PW = 150 ns; $t_r \leq 10$ ns; duty cycle $\leq 2\%$ (see 4.5.3)			V_{F4}			
1N5768, 1N5770, 1N5772, 1N5774, 1N6496, 1N6506, 1N6507, 1N6508, 1N6509		I_F = 500 mA				---	5	V(pk)
1N6100, 1N6101, 1N6510, 1N6511		I_F = 100 mA				---	5	V(pk)
Capacitance (pin-to-pin)	4001	V_R = 0 V; f = 1 MHz (see 4.5.4)			C_t			
1N5770, 1N5772, 1N5774, 1N6496, 1N6507, 1N6508, 1N6509						---	8	pF
1N5768, 1N6100, 1N6101, 1N6506, 1N6510, 1N6511						---	4	pF
Forward recovery time	4026				t_{fr}			
1N5768, 1N5770, 1N5772, 1N5774, 1N6496, 1N6506, 1N6507, 1N6508, 1N6509		I_F = 500 mA dc; R_S = 10Ω; V_{fr} = 1.8 V; $t_r \leq 15$ ns				---	40	ns
1N6100, 1N6101, 1N6510, 1N6511		I_F = 100 mA dc; R_S = 50Ω; V_{fr} = 1.1 V; $t_r \leq 10$ ns				---	15	ns

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	LTPD JANS 1/	LTPD JAN TX 1/ TXV	Symbol	Limits Min	Limits Max	Units
Subgroup 4 - Cont.								
Reverse recovery time	4031				t_{rr}			
1N5768, 1N5770, 1N5772, 1N5774, 1N6496, 1N6506, 1N6507, 1N6508, 1N6509		Condition B; $I_F = I_R = 200$ mA; $R_L = 100\Omega$; $i_{rr} = 20$ mA				---	20	ns
1N6100, 1N6101, 1N6510, 1N6511		Condition B; $I_F = I_R = 10$ mA $R_L = 100\Omega$; $i_{rr} = 1$ mA				---	5	ns
Forward voltage (match)	4011							
1N6100, 1N6101, 1N6510, 1N6511		$I_F = 10$ mA			V_{F5}	---	10	mV
Subgroup 5								
Not applicable								
Subgroup 6								
Surge current	4066	$I_{FSM} = 500$ mA(pk) 10 surges at 1 per minute; $t_p = 1/120$ s; cyclicly choose four diodes from each array.						
Electrical measurements		See table IV, steps 1 and 3.						
Subgroup 7								
Reverse current	4016	DC method; $V_R = 40$ V dc; $I_F = 25$ mA dc for each of the other diodes in the test section. (see 4.5.5)			I_{RX}	---	10	µA dc
1N6100, 1N6101, 1N6510, 1N6511								
Forward voltage	4011	$I_F = 25$ mA dc; $I_F = 25$ mA dc for each of the other diodes in the test section. (see 4.5.5)			V_{FX}	---	1.0	V dc
1N6100, 1N6101, 1N6510, 1N6511								

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See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD		Symbol	Limits		Units
	Method	Conditions	JANS 1/	JAN TX 1/ TXV 1/		Min	Max	
Subgroup 7 - Cont.								
Isolation current	4016	DC method; V_R = 40 V dc (see 4.5.6)			I_{Ri}	---	0.8	µA dc
1N5772								
1N5774								
1N6100								
1N6101								
1N6496								
1N6508								
1N6509								
1N6510								
1N6511								

1/ For LTPD numbers, see MIL-S-19500.

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TABLE IIa. Group B inspection for JANS devices.

Inspection	MIL-STD-750		Qualification 1/ and large lot quality conformance inspection LTPD	Small lot 1/ quality conformance inspection n/c
	Method	Conditions		
Subgroup 1				
Physical dimensions	2066	See figures 1, 2, 3, 4, and 5.		
Subgroup 2				
Solderability	2026			
Resistance to solvents	1022			
Subgroup 3				
Thermal shock (temperature cycling)	1051			
Hermetic seal	1071			
Fine leak				
Gross leak				
Electrical measurements		See table IV, steps 1 and 3.		
Decap internal visual design verification	2075			
Bond strength	2037	Condition A.		
Die shear	2017			
Subgroups 4, 5, 6				
Not applicable				
Subgroup 7			5	N/A
Steady-state operation life	1027	T_A = 150°C each diode DC blocking, V_R = 50 V dc		
Electrical measurements		See table IV, steps 1, 3, 5, and 6.		

See footnote at end of table.

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TABLE IIb. Group B inspection for JAN, JANTX, and JANTXV devices.

Inspection	MIL-STD-750		LTPD 1/
	Method	Conditions	
Subgroup 1			
Solderability	2026		
Resistance to solvents	1022		
Subgroup 2			
Thermal shock (temperature cycling)	1051		
Hermetic seal	1071		
Fine leak			
Gross leak			
Electrical measurements		See table IV, steps 1 and 3.	
Subgroup 3			
Steady-state operation life	1027	DC blocking; T_A = 150°C; V_R = 50 V dc	
Electrical measurements		See table IV, steps 2 and 4.	
Subgroup 4			
Decap internal visual design verification	2075		
Bond strength	2037	Condition A	
Subgroup 5			
Not applicable			
Subgroup 6			
High-temperature life (nonoperating)	1032	T_{STG} = 200°C	
Electrical measurements		See table IV, steps 2 and 4.	

1/ For LTPD number, see MIL-S-19500.

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TABLE III. Group C inspection for (all quality levels).

Inspection	MIL-STD-750		LTPD 1/
	Method	Conditions	
Subgroup 1			
Physical dimensions	2066	See figures 1, 2, 3, 4, and 5.	
Subgroup 2			
Thermal shock (glass strain)	1056	Test condition A	
Terminal strength	2036	Test condition E; 3 oz weight; three bends of 45° for Flatpacks; three bends of 15° for Dips; omit end leads of configuration 2.	
Hermetic seal	1071		
Fine leak			
Gross leak			
Moisture resistance	1021		
Electrical measurements		See table IV, steps 1, 3, 5, and 6 (JANS), and steps 1 and 3 (JAN, JANTX, and JANTXV).	
Subgroup 3			
Shock	2016		
Vibration, variable frequency	2056		
Constant acceleration	2006		
Electrical measurements		See table IV, steps 1, 3, 5, and 6 (JANS), and steps 1 and 3 (JAN, JANTX, and JANTXV).	

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See footnote at end of table.

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TABLE III. Group C inspection for (all quality levels) - Continued.

Inspection	MIL-STD-750		LTPD 1/
	Method	Conditions	
Subgroup 4			
Salt atmosphere (corrosion)	1041		
Subgroup 5			
Not applicable			
Subgroup 6			
Steady-state operation life	1026	T_A= 150°C; V_R = 50 V dc	
Electrical measurements		See table IV, steps 1, 3, 5, and 6 (JANS), and steps 2 and 4 (JAN, JANTX, and JANTXV).	

1/ For LTPD numbers, see MIL-S-19500.

TABLE IV. Groups B and C electrical measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Units
		Method	Conditions		Min	Max	
1.	Reverse current	4016	DC method; V_R = 40 V dc	I_{R1}	---	0.1	µA dc
2.	Reverse current	4016	DC method; V_R = 40 V dc	I_{R1}	---	0.2	µA dc
3.	Forward voltage	4011	I_F = 100 mA dc	V_{F1}	---	1.0	V dc
4.	Forward voltage	4011	I_F = 100 mA dc	V_{F1}	---	1.1	V dc
5.	Reverse current	4016	DC method; V_R = 40 V dc	ΔI_{R1}	---	100% of initial reading or ±20 nA dc, whichever is greater.	
6.	Forward voltage	4011	I_F = 100 mA dc	ΔV_{F1}	---	±25 mV dc of initial reading	

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI, AR, AV
Navy - SH
Air Force - 11, 19, 85
DLA - ES

User activities:
Army - CR, SM
Navy - AS, CG, MC, OS

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5961-1059)

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